

Enzymes

Enzymes

- Enzymes are **protein catalyst** produced by a cell and responsible '**for the high rate**' and **specificity of one or more intracellular or extracellular** biochemical reactions.
- Enzymes are biological catalysts responsible for supporting almost all of the chemical reactions that maintain **animal homeostasis**.
- Almost all enzymes are **globular proteins** consisting either of **a single polypeptide** or of **two or more polypeptides** held together (in quaternary structure) by non-covalent bonds.
- Enzymes do nothing but **speed up the rates** at which the equilibrium positions of reversible reactions are attained.
- In terms of thermodynamics, enzymes **reduce the activation energies of reactions**, enabling them to occur much more readily at low temperatures - essential for biological systems. The basic characteristics of enzymes includes:
 - (i) Almost all the enzymes **are proteins** and they follow the **physical and chemical reactions of proteins**
 - (ii) Enzymes are **sensitive and labile to heat**
 - (iii) Enzymes are **water soluble**
 - (iv) Enzymes could be precipitated by **protein precipitating agents** such as ammonium sulfate and trichloroacetic acid.

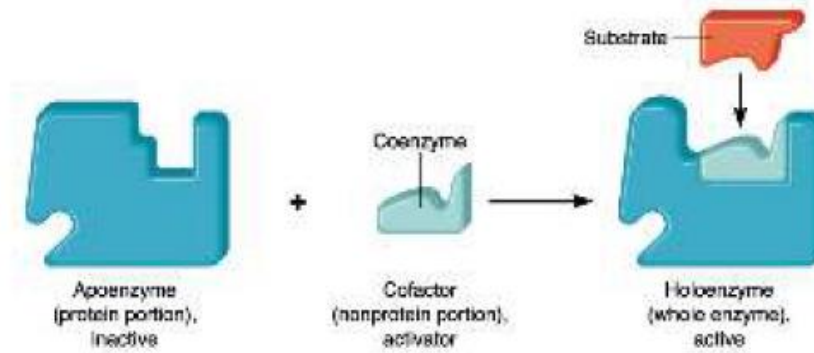
Enzymes

- Catalytic efficiency – high efficiency, 10^3 to 10^{17} faster than the corresponding uncatalyzed reactions
- Specificity - high specificity, interacting with one or a few specific substrates and catalyzing only one type of chemical reaction.
- Mild reaction conditions- 37°C , physiological pH, ambient atmospheric pressure

Enzymes: Composition

(1) Simple protein

(2) Conjugated protein



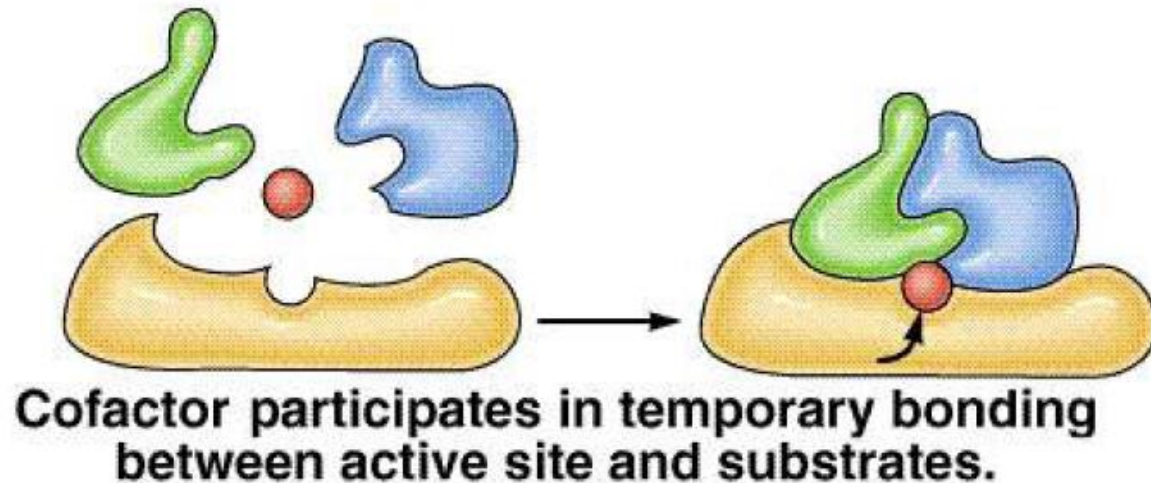
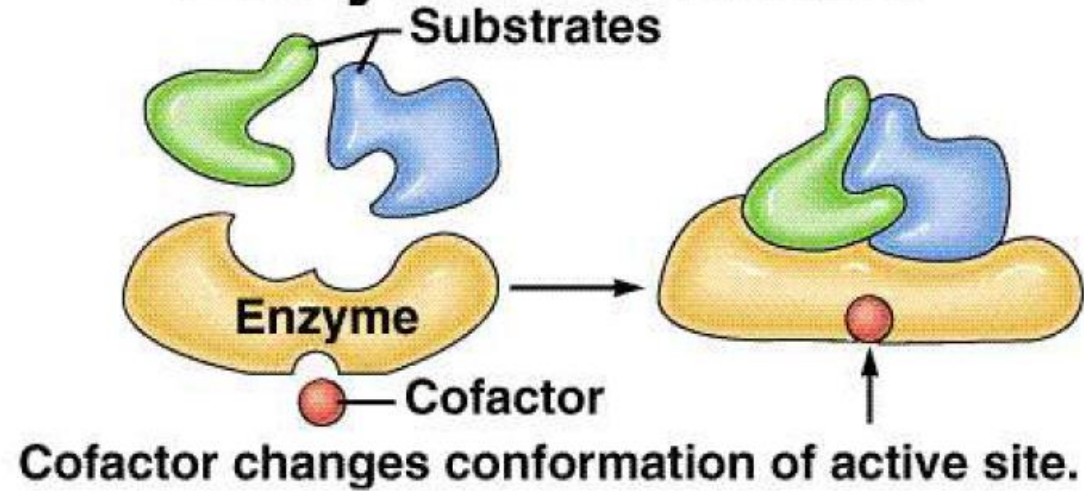
Holoenzyme = Apoenzyme + Cofactor

Cofactor

Coenzyme : loosely bound to enzyme (non-covalently bound).

Prosthetic group : very tightly or even covalently bound to enzyme (covalently bound)

Roles of Cofactors in Enzyme Function



Classification of Enzymes

Since earlier days to still date, fanciful names such as **pepsin**, **chymotrypsin**, etc were used to name enzymes.

Later the suffix “**ase**” to the substrate was used to name enzymes. For example the **enzymes lactase** acts upon the **lactate** and produces **glucose** and **galactose**. The above method is known as “trivial naming” of enzymes.

Currently enzymes are grouped into **six functional classes** by the International Union of Biochemists and Molecular Biology (IUBMB). As per the IUBMB system, each enzyme name starts with EC (enzyme class) followed by 4 digits.

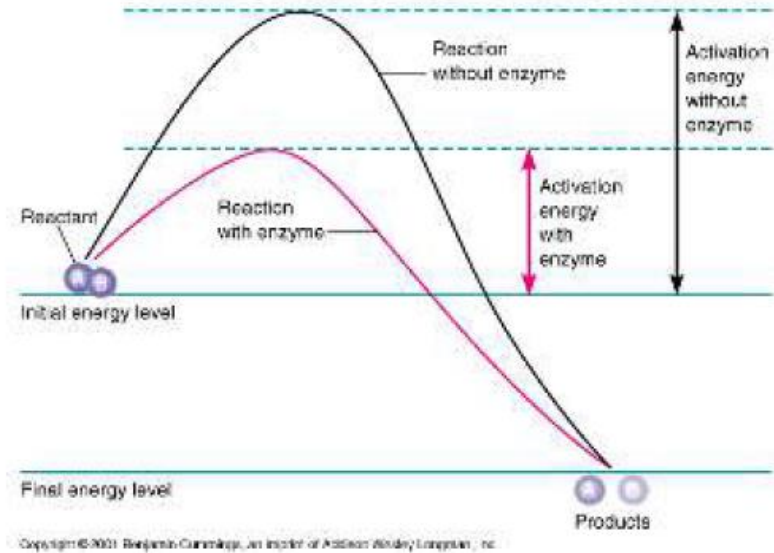
The **first digit represents the class**, the **second digit strands for the subclass**, the **third digit represents the sub-subclass** or subgroup and the fourth digit provides the particular enzyme

Sl.No.	Classification	Biochemical Properties
1.	Oxidoreductases	Act on many chemical groupings to add or remove hydrogen atoms. e.g. Lactate dehydrogenase
2.	Transferases	Transfer functional groups between donor and acceptor molecules. Kinases are specialized transferases that regulate metabolism by transferring phosphate from ATP to other molecules. e.g. Aminotransferase.
3.	Hydrolases	Add water across a bond, hydrolyzing it. E.g. Acetyl choline esterase
4.	Lyases	Add water, ammonia or carbon dioxide across double bonds, or remove these elements to produce double bonds. e.g. Aldolase.
5.	Isomerases	Carry out many kinds of isomerization: L to D isomerizations, mutase reactions (shifts of chemical groups) and others. e.g. Triose phosphate isomerase
6.	Ligases	Catalyze reactions in which two chemical groups are joined (or ligated) with the use of energy from ATP. e.g. Acetyl CoA carboxylase

How Enzymes work?

1) Enzymes lower a reaction's activation energy

- All chemical reactions have an energy barrier, called the **activation energy**, separating the reactants and the products.
- **activation energy:**
amount of energy needed to disrupt stable molecule so that reaction can take place.



How Enzymes work?

- Enzymes bind substrates to their active site and stabilize the transition state of the reaction.
- The **active site** of the enzyme is the place where the substrate binds and at which catalysis occurs.
- The active site binds the substrate, forming an enzyme-substrate(ES) complex.

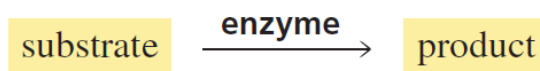


CATALYSIS IN BIOLOGICAL REACTIONS

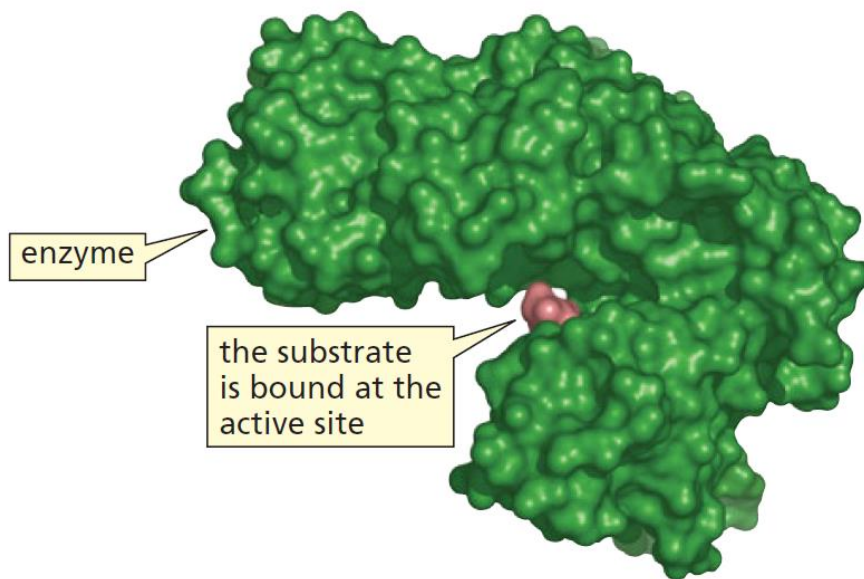
Essentially all organic reactions that occur in cells require a catalyst. Most biological catalysts are **enzymes**, which are globular proteins. Each biological reaction is catalyzed by a different enzyme.

Binding the Substrate

The reactant of an enzyme-catalyzed reaction is called a **substrate**.

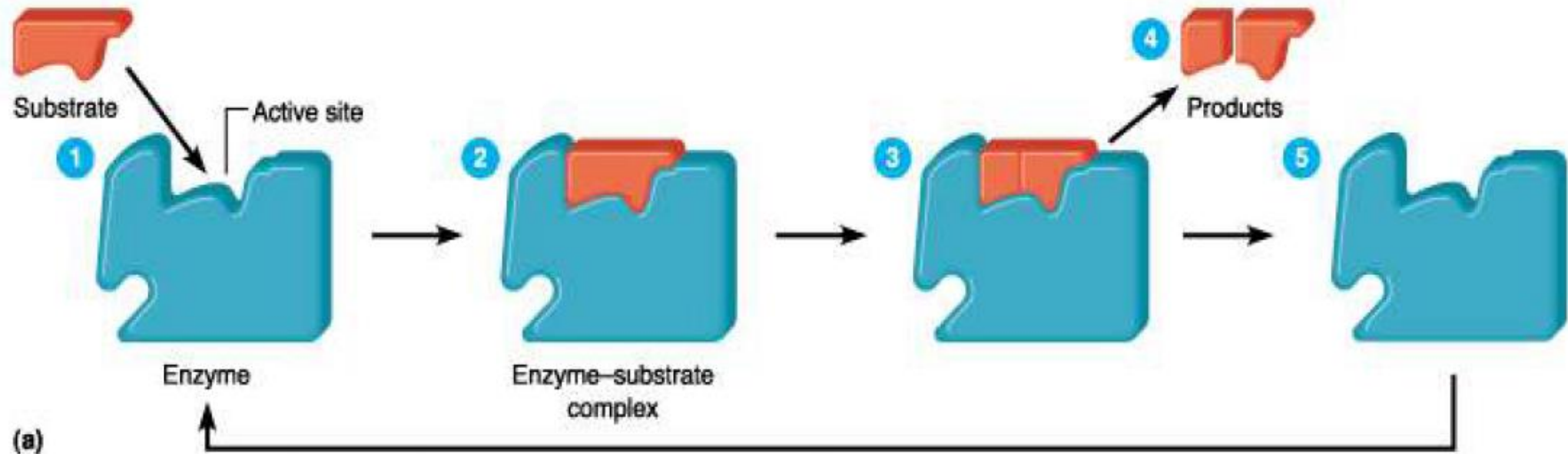


The enzyme binds its substrate in its **active site**, a pocket in the cleft of the enzyme. All the bond-making and bond-breaking steps that convert the substrate to the product occur while the substrate is bound to the active site.



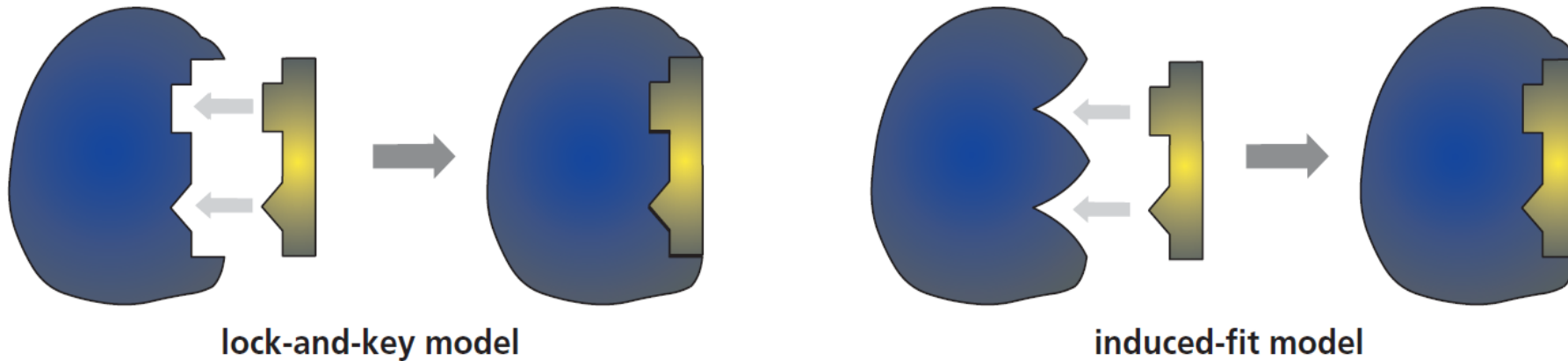
CATALYSIS IN BIOLOGICAL REACTIONS

- Enzymes differ from **nonbiological catalysts** in that they are **specific for the substrate** whose reaction they catalyze.
- Enzymes, however, have **different degrees of specificity**. Some enzymes are **specific** for a single compound. For example, **glucose- 6-phosphate isomerase** catalyzes the isomerization of glucose-6-phosphate only.
- On the other hand, some enzymes catalyze the **reactions of several compounds with similar structures**. For instance, **hexokinase** catalyzes the phosphorylation of **any D-hexose**.
- The specificity of an enzyme for its substrate is another example of **molecular recognition** — the ability of one molecule to recognize another molecule.



CATALYSIS IN BIOLOGICAL REACTIONS

In 1894, Emil Fischer proposed the **lock-and-key model** to account for the specificity of an enzyme for its substrate. This model related the specificity of an enzyme for its substrate to the specificity of a lock for a correctly shaped key.



In 1958, Daniel Koshland proposed the **induced-fit model** of substrate binding. In this model, the shape of the active site does not become completely complementary to the shape of the substrate until the enzyme has bound the substrate. The energy released as a result of binding the substrate can be used to induce a change in the conformation of the enzyme, leading to more precise binding between the substrate and the active site. An example of induced fit is shown in Figure

Nucleic Acids

Nucleic Acids

Nucleic acid, naturally occurring chemical compound that is capable of being broken down to yield **phosphoric acid, sugars, and a mixture of organic bases** (**purines** and **pyrimidines**). Nucleic acids are the main information-carrying molecules of the cell, and, by directing the process of protein synthesis, they determine the inherited characteristics of every living thing. The two main classes of nucleic acids are **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**.

DNA was first isolated in 1869 from the **nuclei of white blood cells**. Because it was found in the **nucleus** and **was acidic**, it was called ***nucleic acid***. Eventually, scientists found that the nuclei of all cells contain DNA, but not until it was shown in 1944 that DNA **could be transferred from one species to another**, along with inheritable traits, did they realize that DNA is the carrier of genetic information. In 1953, **James Watson and Francis Crick** described the **three-dimensional structure of DNA—the famed double helix**.

DNA is the master blueprint for **life and constitutes the genetic material in all free-living organisms** and most viruses. **RNA is the genetic material of certain viruses**, but it is also found in all living cells, where it plays an important role in certain processes such as the making of proteins.

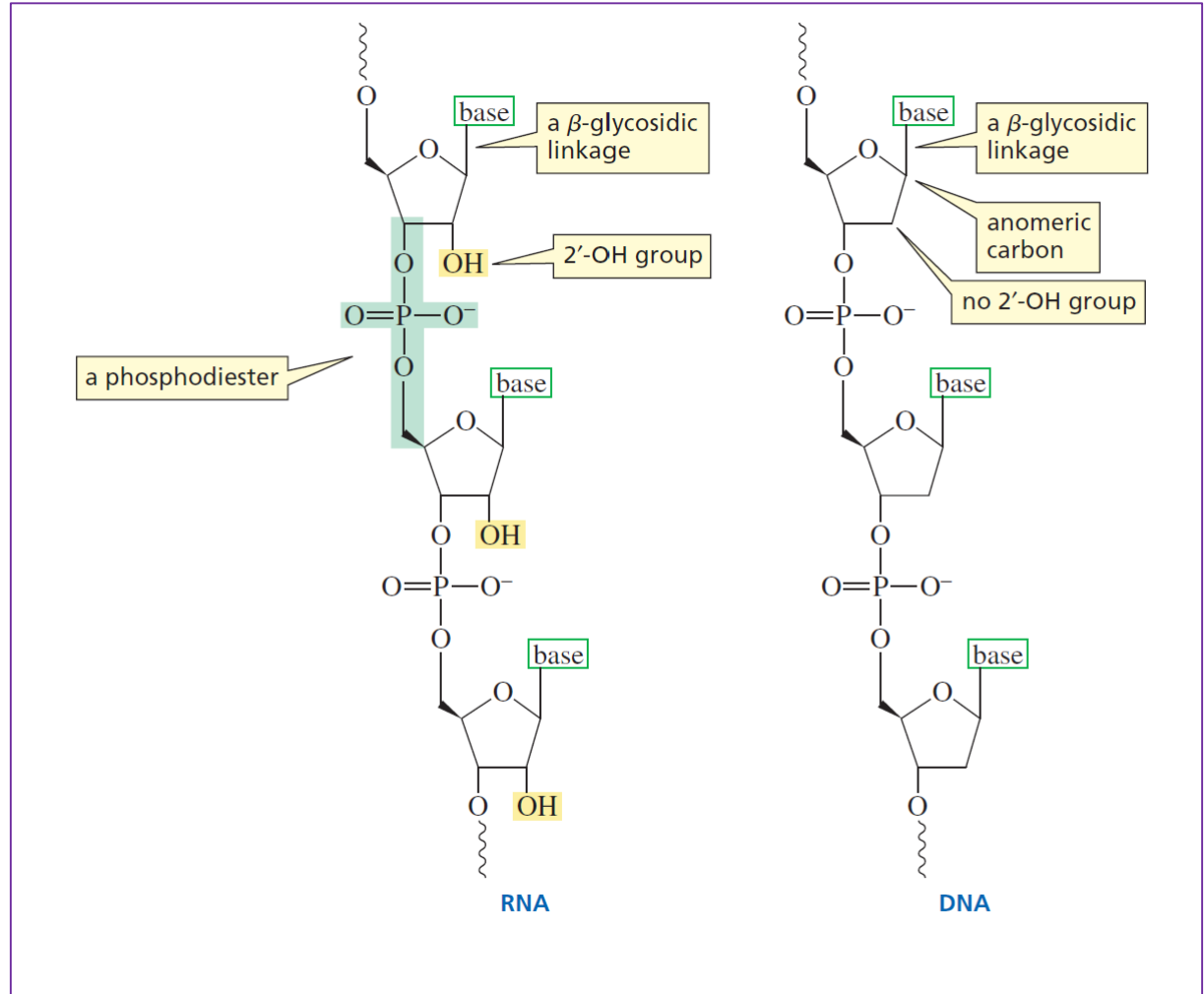
NUCLEOSIDES AND NUCLEOTIDES

Nucleic acids are chains of five-membered-ring sugars linked by phosphate groups.

Nucleoside = base + sugar

Nucleotide = base + sugar + phosphate

- Notice that the linkages are **phosphodiester**s.
- In **RNA**, the five-membered-ring sugar is **D-ribose**. In DNA it is **2-deoxy-D-ribose** (D-ribose without an OH group in the 2-position).
- The anomeric carbon of each sugar is bonded **to a nitrogen of a heterocyclic compound** in a **β -glycosidic linkage**.
- A **β -linkage** is one in which the substituents at C-1 and C-4 are on the same side of the furanose ring.
- Because the **heterocyclic compounds are amines**, they are commonly referred to as **bases**.

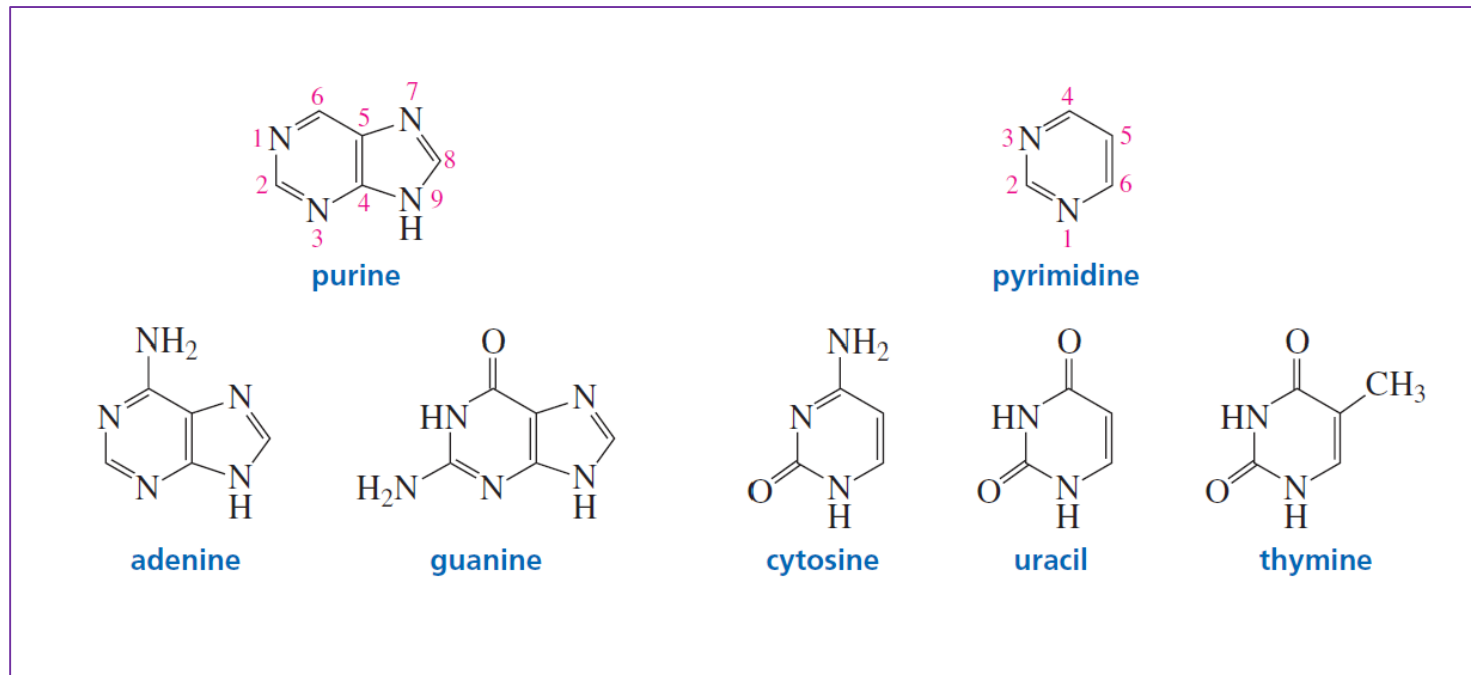


NUCLEOSIDES AND NUCLEOTIDES

The vast differences in heredity between different species and between different members of the same species are determined by the **sequence of the bases in DNA**.

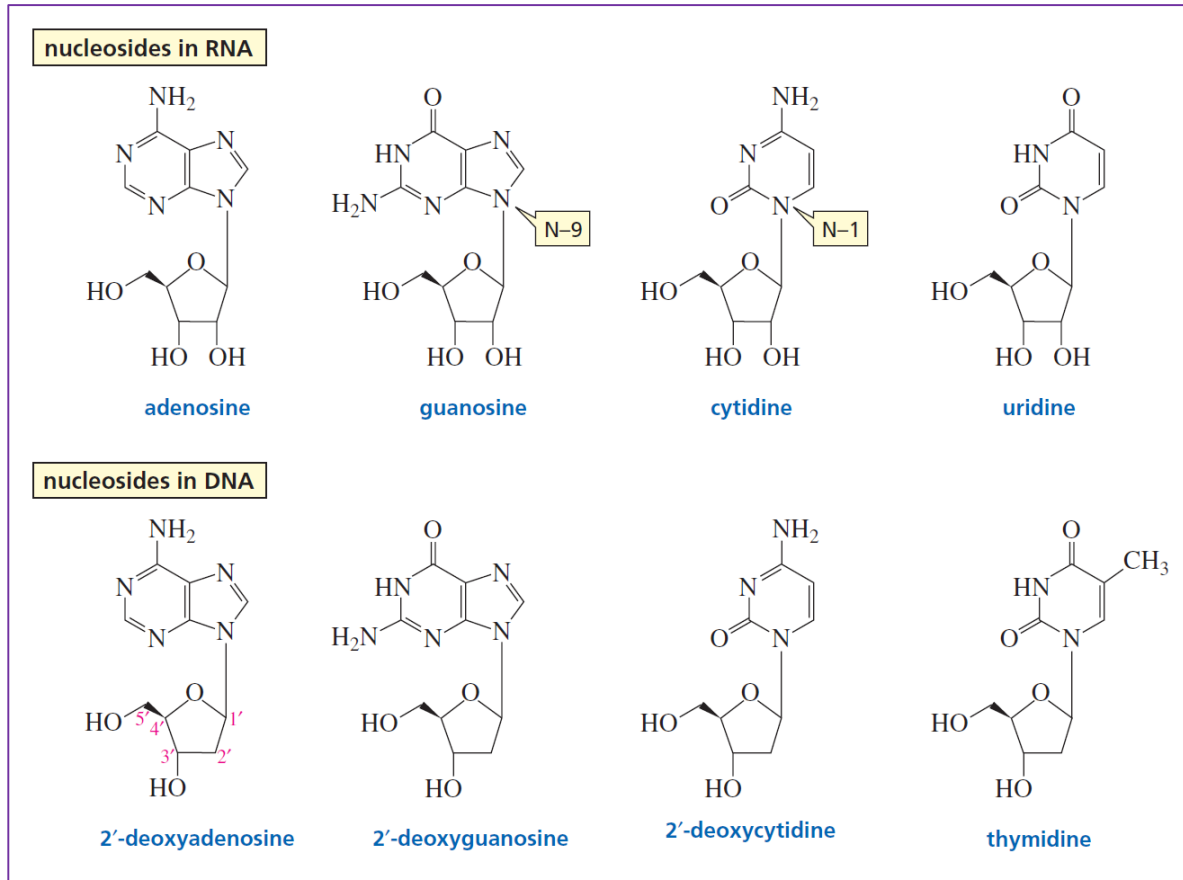
There are only four bases in **DNA**: two are substituted **purines** (*adenine* and *guanine*), and two are substituted **pyrimidines** (*cytosine* and *thymine*).

RNA also contains only four bases. Three (*adenine*, *guanine*, and *cytosine*) are the same as those in DNA, but the fourth base in RNA is **uracil** instead of *thymine*.



NUCLEOSIDES AND NUCLEOTIDES

- The anomeric carbon of the furanose ring is bonded to **purines at N-9** and to **pyrimidines at N-1**. A compound containing a base bonded to d -ribose or to 2-deoxy- d-ribose is called a **nucleoside**.
- The ring positions of the sugar component of a nucleoside are indicated by primed numbers to distinguish them from the ring positions of the base. This is why the sugar component of **DNA** is referred to as **2-deoxy- D -ribose**.
- The nucleosides of **RNA**— where the sugar is **D-ribose**— are more precisely called **ribonucleosides**, whereas the nucleosides of **DNA**—where the sugar is **2-deoxy-d-ribose**—are called **deoxyribonucleosides**.

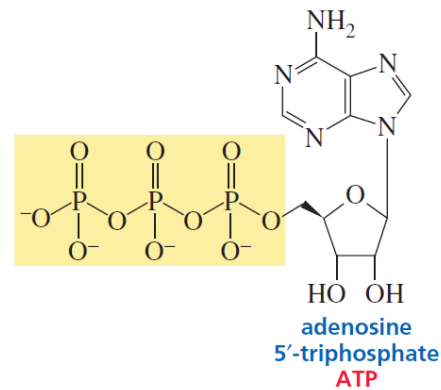
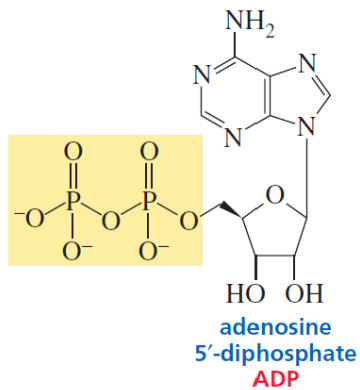
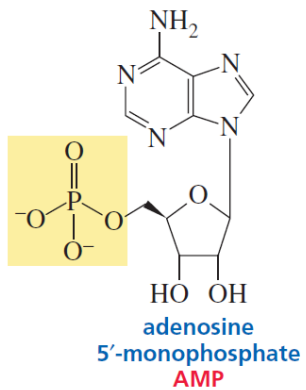


NUCLEOSIDES AND NUCLEOTIDES

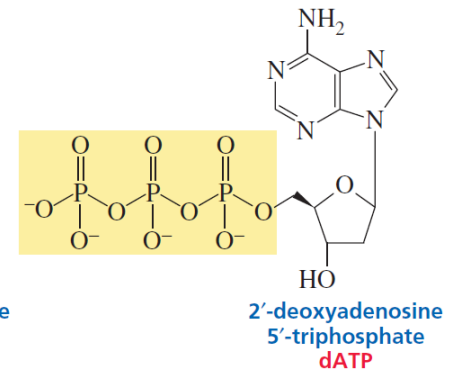
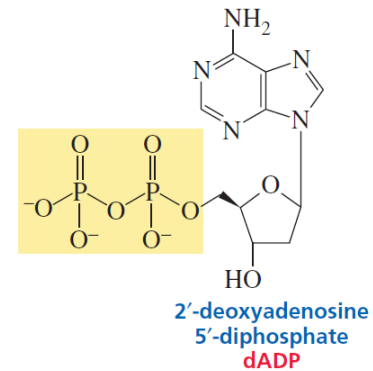
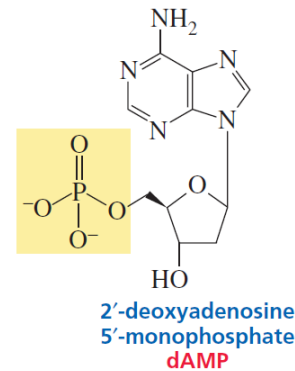
A **nucleotide** is a nucleoside with an OH group of the sugar bonded in an ester linkage to phosphoric acid. The nucleotides of RNA are more precisely called **ribonucleotides**, and those of DNA are called **deoxyribonucleotides**.

Because **phosphoric acid** can form **an anhydride**, **nucleotides** can exist as **monophosphates, diphosphates, and triphosphates**. They are named by adding *monophosphate* or *diphosphate* or *triphosphate* to the name of the nucleoside.

nucleosides of adenosine



nucleosides of 2'-deoxyadenosine

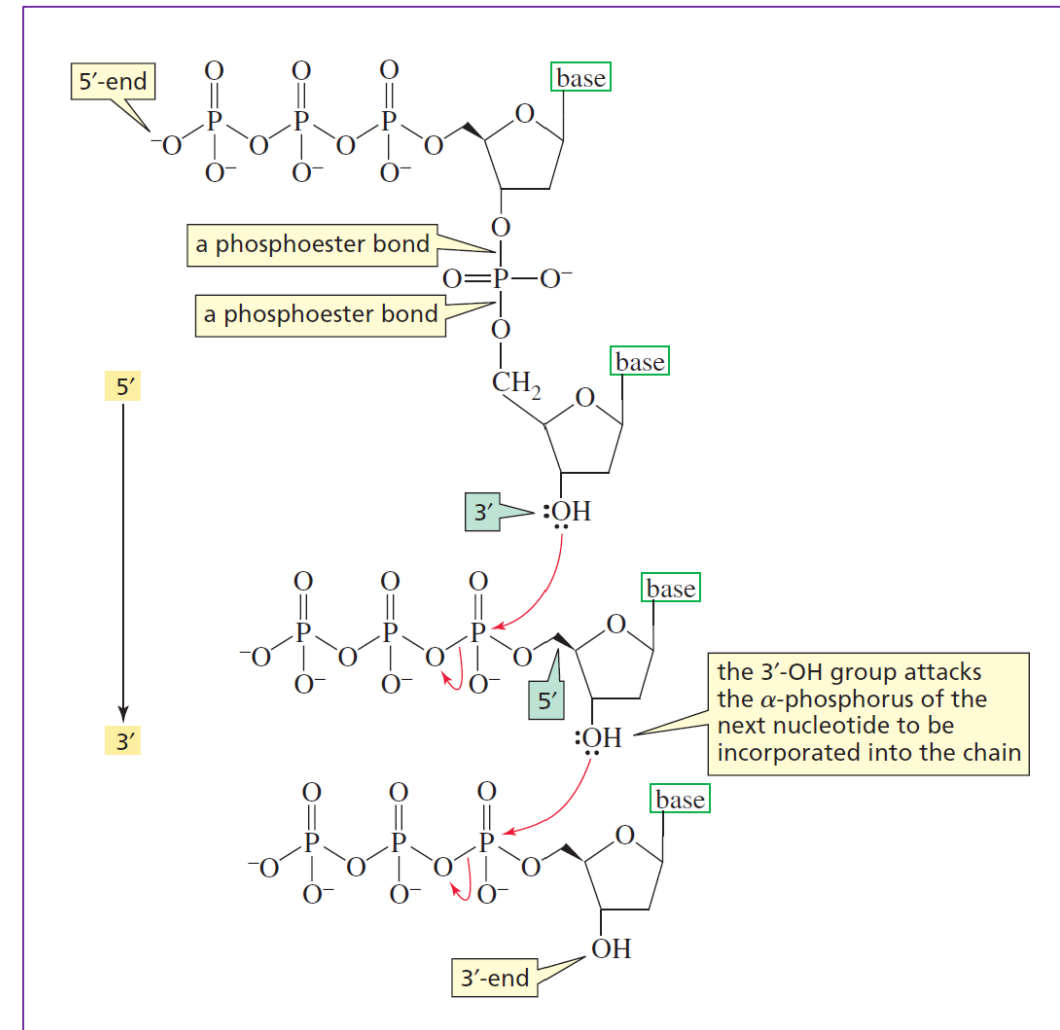


NUCLEIC ACIDS ARE COMPOSED OF NUCLEOTIDE SUBUNITS

- Nucleic acids are composed of long strands of nucleotide subunits.
- A **dinucleotide** contains two nucleotide subunits, an **oligonucleotide** contains 3 to 10 subunits, and a **polynucleotide** contains many subunits. DNA and RNA are **polynucleotides**.

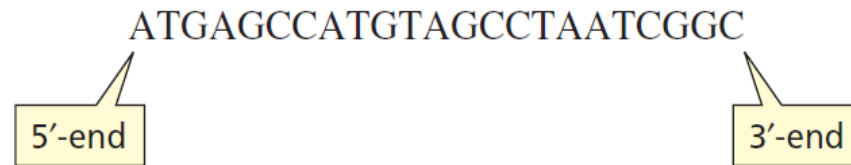
Nucleotide triphosphates are the starting materials for the **biosynthesis of nucleic acids**. DNA is synthesized by enzymes called **DNA polymerases**; RNA is synthesized by enzymes called **RNA polymerases**. The nucleotides are linked as a result of nucleophilic attack by a **3-OH group** of one nucleotide triphosphate on the **α -phosphorus** of another **nucleotide triphosphate**, breaking a phosphoanhydride bond and eliminating pyrophosphate.

Thus, the phosphodiester joins the **3'-OH** group of one nucleotide and the **5'-OH** group of the next nucleotide, and the growing polymer is synthesized in the **5'→3' direction**. In other words, new nucleotides **are added to the 3-end**. **Pyrophosphate is subsequently hydrolyzed**, which makes the reaction **irreversible**. Irreversibility is important if the **genetic information in DNA is to be preserved**. RNA strands are biosynthesized in the same way, using ribonucleotides instead of 2-deoxyribonucleotides.



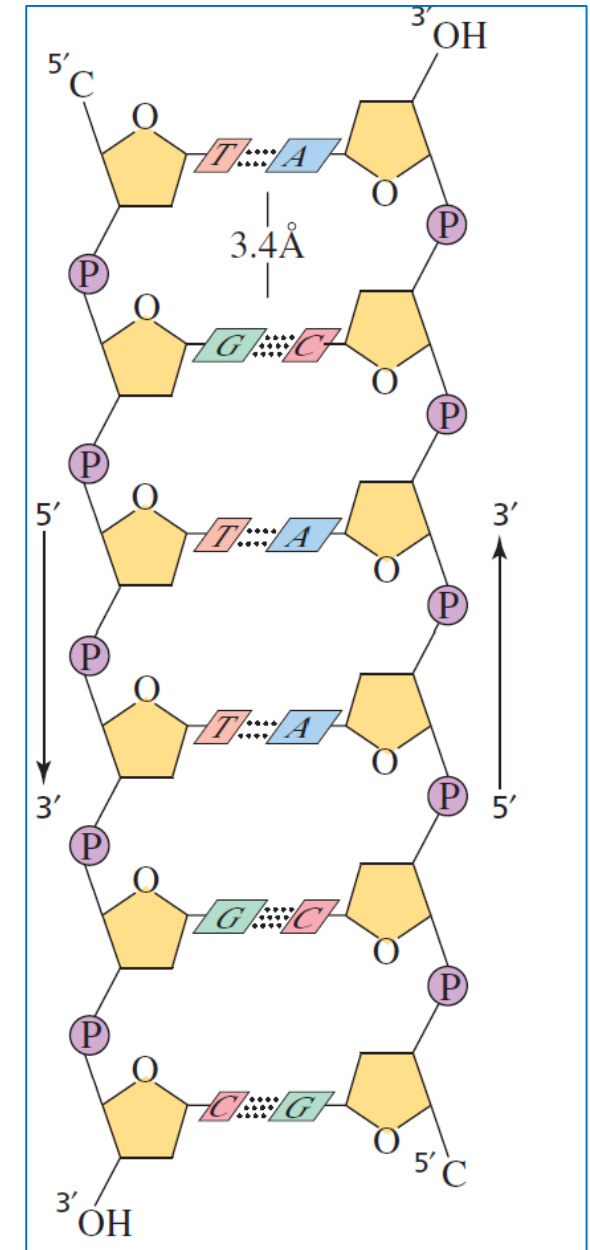
NUCLEIC ACIDS ARE COMPOSED OF NUCLEOTIDE SUBUNITS

The **primary structure** of a nucleic acid is the sequence of bases in the strand. By convention, the sequence of bases is written in the 5' → 3' direction (the 5'-end is on the left). Remember that the nucleotide at the 5'-end of the strand has an unlinked 5'-triphosphate group, and the nucleotide at the 3'-end has an unlinked 3'-hydroxyl group.



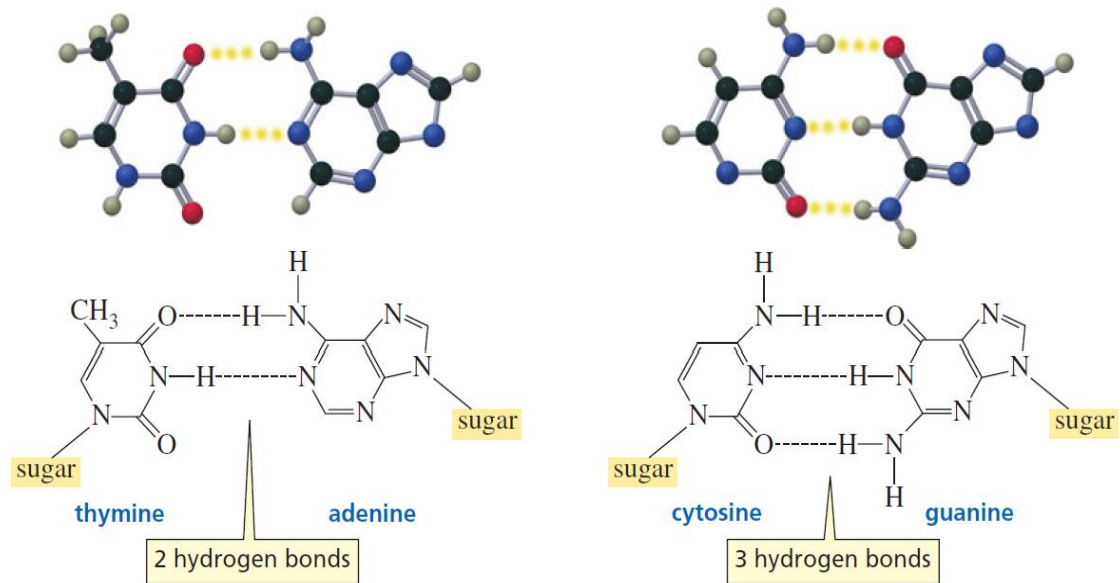
Watson and Crick concluded that DNA consists of two strands of nucleotides, with the sugar–phosphate backbone on the outside and the bases on the inside. The strands are antiparallel (they run in opposite directions) and are held together by hydrogen bonds between the bases on one strand and the bases on the other strand.

Experiments carried out by Erwin Chargaff were critical to Watson and Crick's proposal for the structure of DNA. These experiments showed that **the number of adenines in DNA equals the number of thymines, and the number of guanines equals the number of cytosines.**



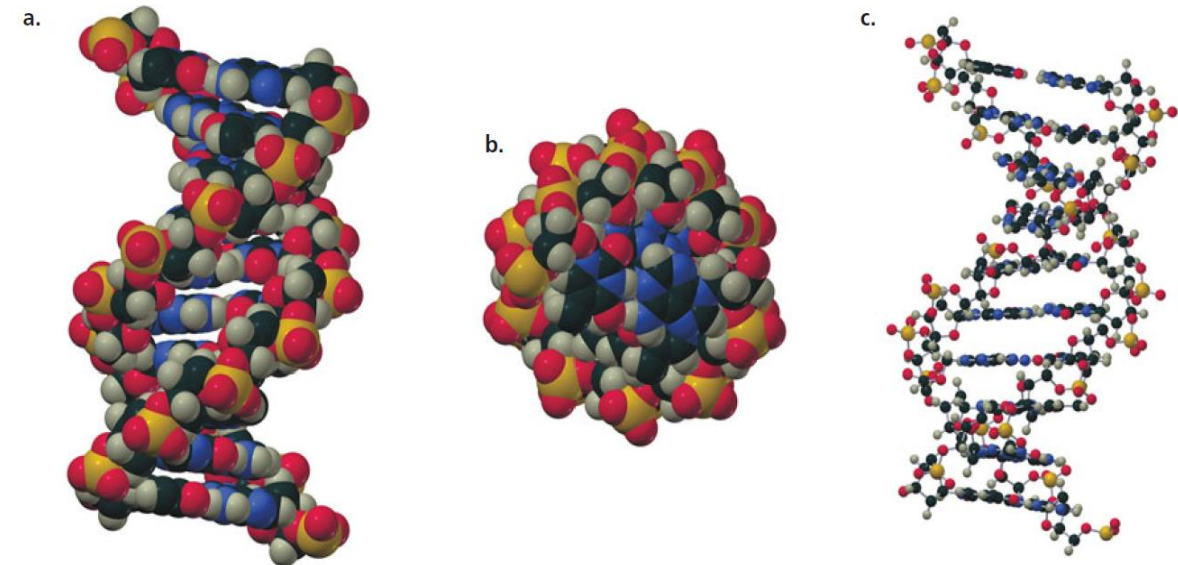
DNA-STRUCTURE

The base pairing is dictated by **hydrogen bonding**. Adenine forms **two hydrogen bonds with thymine** but would form only **one hydrogen bond with cytosine**. Guanine forms **three hydrogen bonds with cytosine** but would form **only one hydrogen bond with thymine**.



▲ Base pairing in DNA: adenine and thymine form two hydrogen bonds; cytosine and guanine form three hydrogen bonds.

The two antiparallel DNA strands are not linear but are **twisted into a helix around a common axis**. The base pairs are **planar and parallel to each other on the inside of the helix**. The secondary structure is therefore known as a **double helix**. The double helix resembles a circular staircase: the base pairs are the **rungs**, and the **sugar-phosphate backbones are the handrails**.



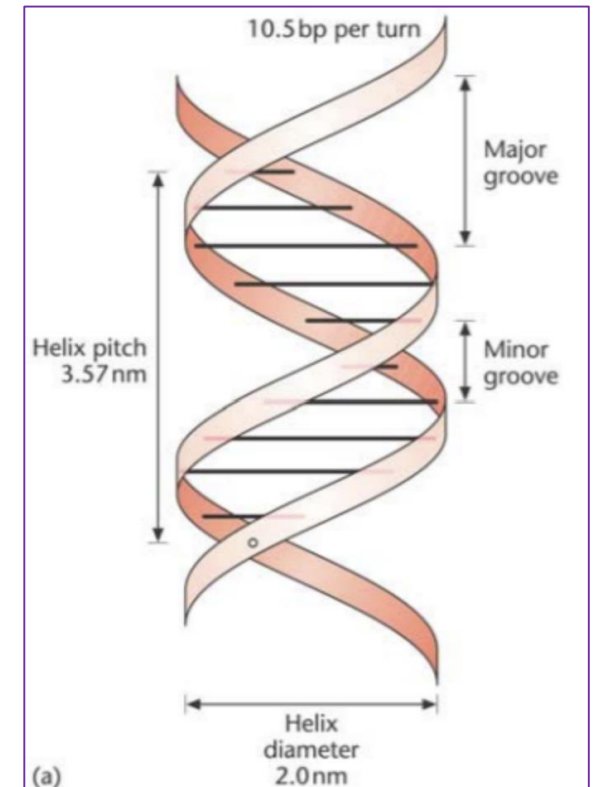
- ▲
- a. The DNA double helix.
 - b. The view looking down the long axis of the helix.
 - c. The bases are planar and parallel on the inside of the helix.

DNA-STRUCTURE

Hydrogen bonding between base pairs is just one of the forces holding the two strands of the **DNA double helix** together. The bases are planar aromatic molecules that stack on top of one another, each pair slightly rotated with respect to the next pair, like a partially spread-out hand of cards. In this arrangement, there are favorable van der Waals interactions between the **mutually induced dipoles of adjacent pairs of bases**. These interactions, known as **stacking interactions**, are weak attractive forces, but when added together **they contribute significantly** to the stability of the double helix. Stacking interactions are **strongest between two purines** and **weakest between two pyrimidines**. Confinement of the bases to the inside of the helix has an additional stabilizing **effect—it reduces the surface area of the relatively nonpolar residues** that is exposed to water, which increases the entropy of the surrounding water molecules.

There are two different alternating grooves in a DNA helix; a **major groove** and a narrower **minor groove**.

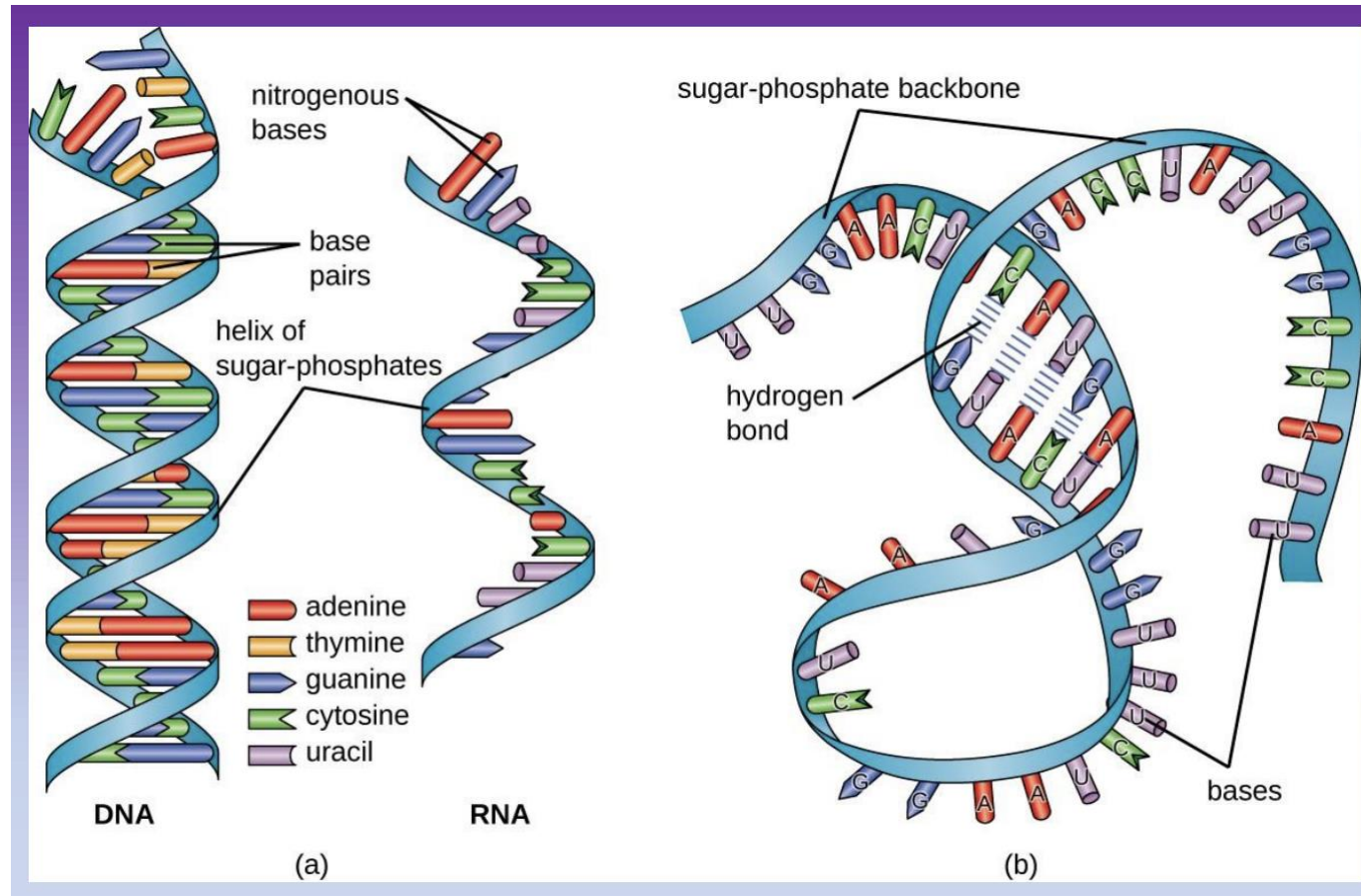
Proteins and other molecules **can bind to the grooves**. The hydrogen-bonding properties of the functional groups **facing into each groove determine** what kind of molecules will bind to the groove.



RNA-STRUCTURE

- RNA is typically **single stranded**.
- Is made of ribonucleotides that are linked by **phosphodiester bonds**.
- A ribonucleotide in the RNA chain contains ribose (the pentose sugar), one of the four nitrogenous bases (A, U, G, and C), and a phosphate group.
- The subtle structural difference between the sugars **gives DNA added stability, making DNA more suitable for storage of genetic information**, whereas the **relative instability of RNA makes it more suitable for its more short-term functions**.
- The RNA-specific pyrimidine **uracil** forms a complementary base pair with **adenine** and is used **instead of the thymine** used in DNA.
- Even though RNA is single stranded, most types of RNA molecules show extensive **intramolecular base pairing** between complementary sequences within the RNA strand, creating a predictable three-dimensional structure essential for their function.

RNA-STRUCTURE



- a) DNA is typically double stranded, whereas RNA is **typically single stranded**.
- b) Although it is single stranded, RNA can fold upon itself, with the folds stabilized by short areas of complementary base pairing within the molecule, forming a **three-dimensional structure**.

LIPIDS

Lipids

Lipid is the collective name for fats, oils, waxes and fat-like molecules (such as steroids) found in the body. Their roles include:

- components of cell membranes (phospholipids and cholesterol)
- energy stores
- chemical messengers (steroid 'hormones')
- protection, waterproofing, insulation and buoyancy agents.

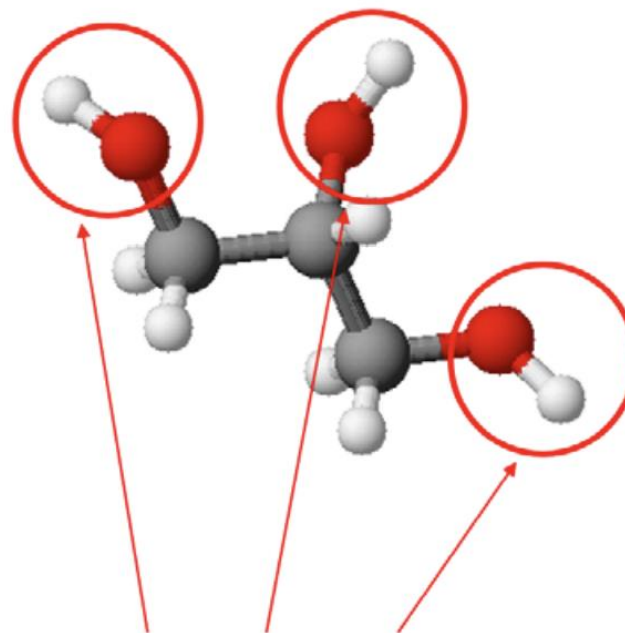
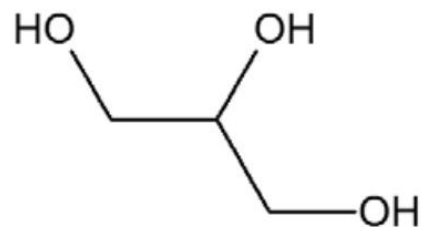
The basic unit of lipids is a triglyceride, synthesised from glycerol (propane-1,2,3-triol) and fatty acids.

Glycerol is a type of alcohol. **Alcohols** are organic compounds. Their molecules are characterised by **hydroxyl groups**, -OH. Glycerol is a **trihydric** alcohol, which means each molecule has three hydroxyl groups.

Glycerol, $C_3H_8O_3$

Glycerol is a trihydric alcohol, in other words each molecule has three hydroxyl groups

Its skeletal formula is



There is one hydroxyl (alcohol) group attached to each carbon atom in the chain

Fatty acids

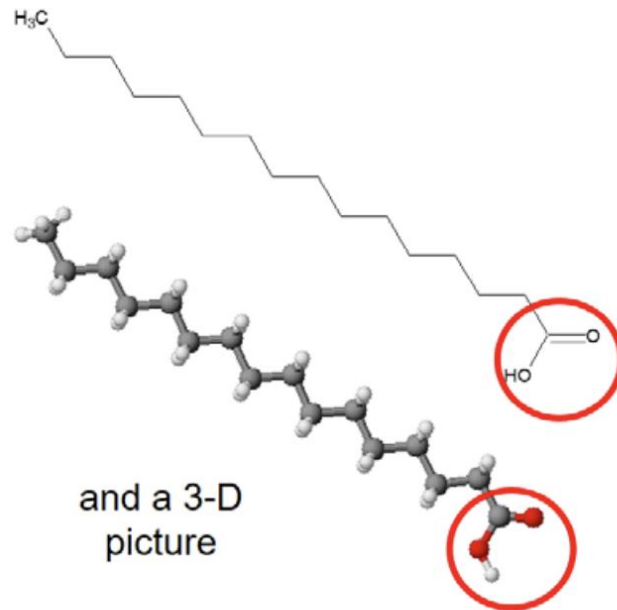
Fatty acids are also organic compounds. Each fatty acid has a **carboxylic acid** (-COOH) group. Some fatty acids are saturated. These means all carbon-carbon bonds are single covalent bonds. Other fatty acids are unsaturated. This means that while most carbon-carbon bonds are single covalent bonds, some are double bonds.

Fatty acids

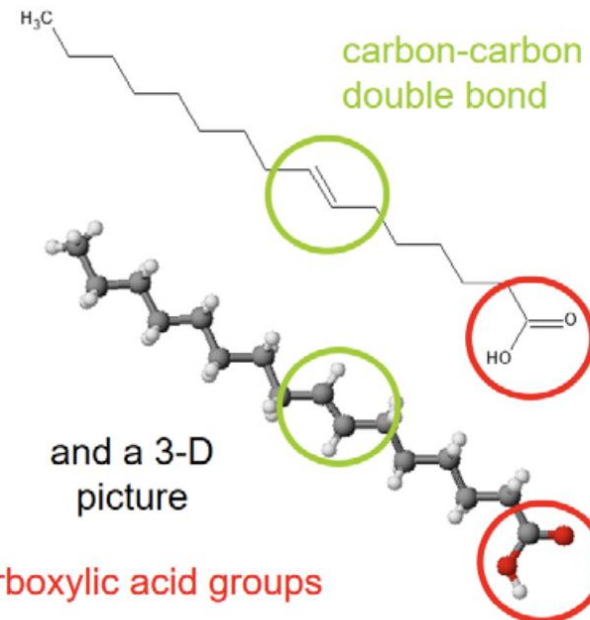
Fatty acids are carboxylic acids with a long hydrocarbon tail

They may be **saturated** or **unsaturated**

Here is the skeletal formula for a typical **saturated** fatty acid

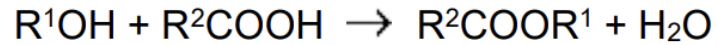


Here is the skeletal formula for a typical **unsaturated** fatty acid



Esters

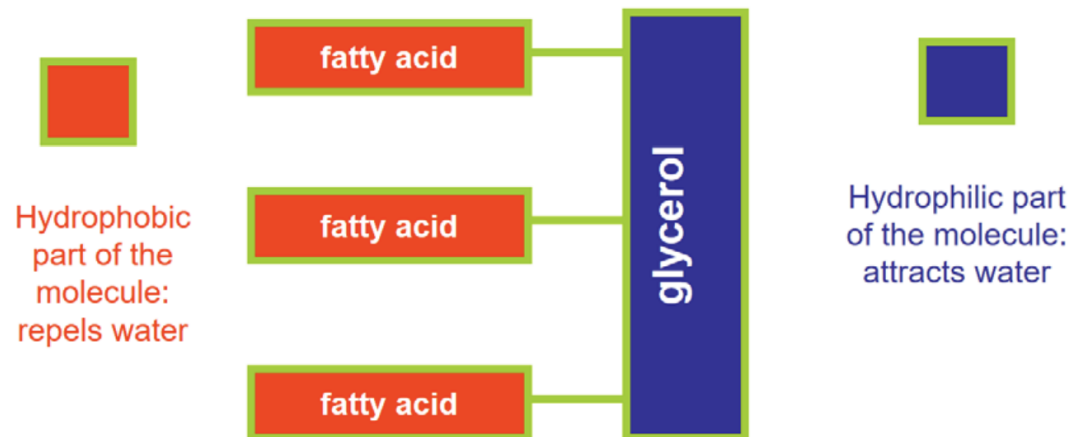
Alcohols react with carboxylic acids to form esters



As with the formation of glycosidic and peptide bonds, this is a **condensation reaction**. A water molecule splits out. The glycerol molecule has three hydroxyl groups. When one reacts with a fatty acid a monoglyceride forms.

When two react with two fatty acid molecules a diglyceride forms. And when three react with three fatty acid molecules a triglyceride forms.

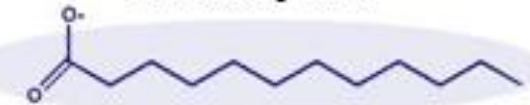
Representation of a triglyceride



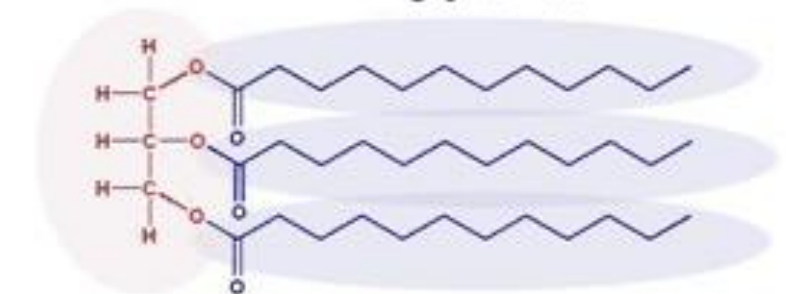
Glycerol



Free fatty acid



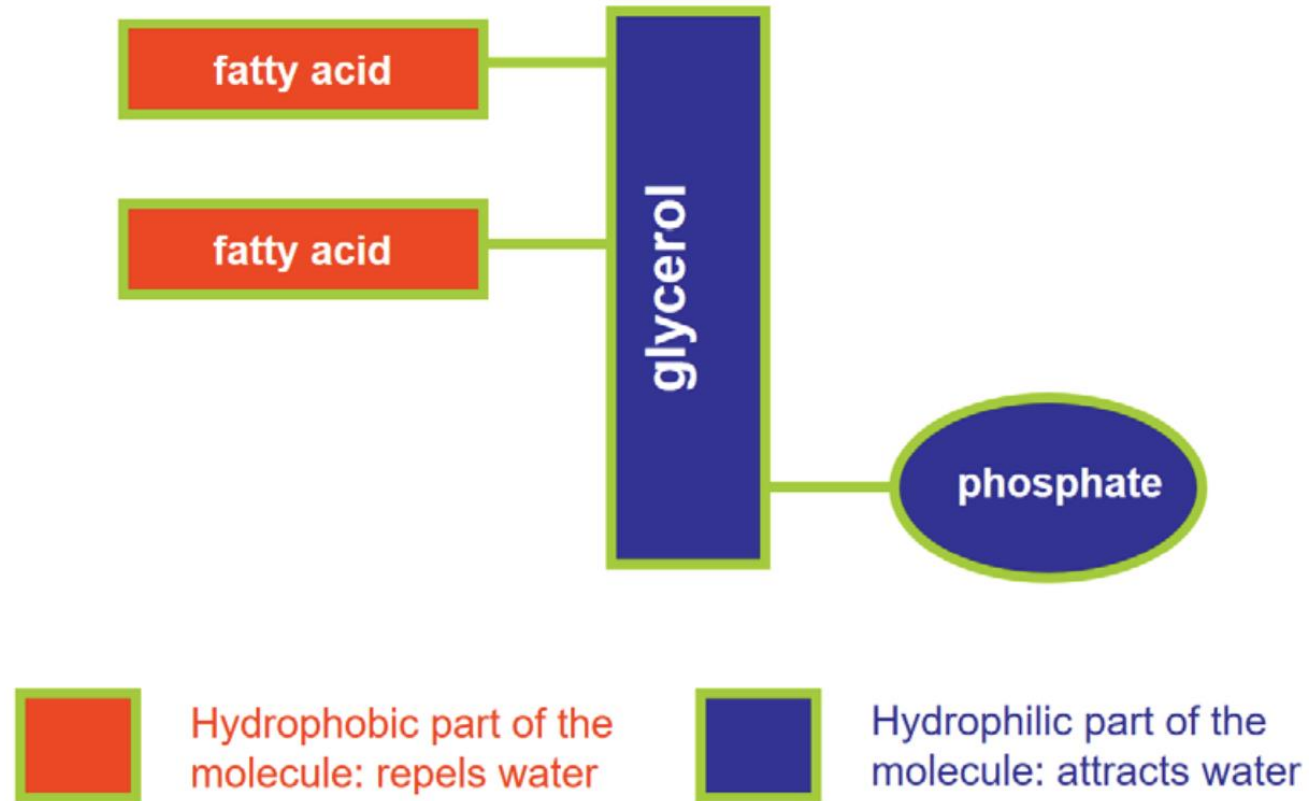
Triglyceride



Phospholipids

Phosphoric acid is an inorganic compound. Its molecular formula is H_3PO_4 . Like fatty acids, it can react with alcohols to form esters (in this case, usually called phosphate esters). In phospholipids, one of the fatty acids of a triglyceride is substituted by a phosphate group.

Representation of a phospholipid



Phospholipids have a key role in the structure and properties of **cell membranes**.